QUANT

Averages, LCM & Decimals

Assignment 04

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1. In the month of July of a certain year, the average daily expenditure of an organization was Rs. 68. For the first 15 days of the month, the average daily expenditure was Rs. 85 and for the last 17 days, Rs. 51. Find the amount spent by the organization on the 15th of the month.

1. Rs. 42 2. Rs. 36 3. Rs. 34 4. Rs. 52 5. Rs. 43

2. The average weight of 5 men is decreased by 3 kg when one of them weighing 150 kg is replaced by another person. This new person is again replaced by another person whose weight is 30 kg lower than the person he replaced. What is the overall change in the average due to this dual change?

1. 6 kg 2. 9 kg 3. 12 kg 4. 15 kg 5. 3 kg.

3. 19 persons went to a hotel for a combined dinner party. 13 of them spent Rs. 79 each on their dinner and the rest spent Rs. 4 more than the average expenditure of all the 19. What was the approximate total money spent by them (in Rs.)?

1. 1,518.4 2. 1,434 3. 1,492 4. 1502 5. 1539

4. The average of 'n' numbers is z. If the number x is replaced by the number x^1, then the average becomes z^1. Find the relation between n, z, z^1, x and x^1.

1. $\frac{z^1-2}{x^1-x}=\frac{1}{n}$ 2. $\frac{x^1-x}{z^1}=\frac{1}{n}$ 3. $\frac{z-z^1}{x-x^1}=\frac{1}{n}$ 4. $\frac{x-x^1}{z-z^1}=\frac{1}{n}$ 5. $\frac{z-z^i}{x-x^1}=n$

5. A team of 7 students with average weight 132 kg. is combined with another team of 5 students with average weight of 128 kg. What is the average weight of the new team?

1. 130.34 kg. 2. 132.68 kg. 3. 134.42 kg. 4. 129.6 kg. 5. 162.8 kg.

6. The average temperature for Monday, Tuesday, Wednesday and Thursday is 56 degrees. The average temperature for Tuesday, Wednesday, Thursday and Friday is 61 degrees. If the ratio of the temperatures on Monday and Friday is 5 : 9, find the temperature on Friday.

1. 25 degrees 2. 40 degrees 3. 45 degrees 4. 38 degrees 5. 40 degrees

7. The average age of husband and wife was 23 years, 5 years ago when they were married. At present the average age of the husband, the wife and a child is 20 years. How many years after the marriage was the child born?

1. 2 yrs 2. 4 yrs 3. 3 yrs 4. 6 yrs 5. 1 yr

8. What is the average of first 'n' whole numbers?

1. $\frac{n-1}{2}$ 2. $\frac{n}{2}$ 3. $\frac{n+1}{2}$ 4. $\frac{n}{3}$ 5. Data inadequate

9. A submarine can travel at 24 knots on the surface and at 16 knots when submerged. Find its average speeds if it is submerged (i) for three quarters of the distance and (ii) for three quarters of the time.

1. 16.45, 18 2. 17.45, 19 3. 16.45, 20 4. 17.45, 18 5. Data inadequate

10. The average weight of 40 passengers in a bus is 50 kg., while the average weight of passengers along with its driver and the conductor is 51 kg. Hence the average weight of the driver and the conductor must be ____.

1. 71 kg. 2. 70 kg. 3. 72 kg. 4. 74 kg. 5. 81 kg.

Answer Key & Explanations

Test - 04	
1.	3
2.	2
3.	5
4.	3
5.	1
6.	3
7.	4
8.	2
9.	4
10.	1

Test - 04

1. Standard question; requires good calculation speed. Obviously, the 15th day is being double counted. Calculations can be reduced by thinking as: Surplus in first 15 days – Deficit in last 17 days = 255 – 289
→ Net deficit of 34. This means that the average is reducing by 34 due to the double counting of the 15th day. This can only mean that the 15th day's expenditure is Rs. 34 less than the average i.e. Rs. 68 – 34 = 34.

2. The weight of the second man is 135 and that of the third is 105. Hence, net result is a drop of 45 for 5 people. Hence, 9 kg is the drop.

3. Assume x is the average expenditure of 19 people. Then, 19 x = 13 ×79 + 6 (x + 4) ⇒ 13x = 1051 ⇒ x ≅ 81. Thus the total expenditure is 81 × 19 = 1539.

4. $nz - x + x^1 = nz^1$ → simplify to get option 3rd.

5. Just make the equation and get the answer as
$\frac{[7 \times 132 + 5 \times 128]}{12} = 130.34.$

6. $\frac{M+T+W+TH}{4} = 56 \Rightarrow$ M + T + W + TH = 224.
$\frac{T+W+TH+F}{4} = 61 \Rightarrow$ T + W + TH + F = 244
⇒ F – M = 20. Also F: M = 9: 5 ⇒ 9x and 5x
∴ 9x -5x = 20 ⇒ x = 5
∴ Temperature on Friday is 9x = $45^{0.}$

7. $\frac{(H+W)}{2}$ =23 (5 years ago) ∴ Their total age 5 years ago = 46. Hence their total present age = 46 + 5 + 5 = 56 yrs. Total age of their family (H + W + C) = 20 × 3 = 60 yrs. ∴Age of child = 60 – 56 = 4 yrs.
Hence child was born 5 – 4 = 1 year after marriage.

8. n whole numbers means (n – 1) natural numbers because 0 is not a natural number. Sum of 1st (n – 1) natural numbers = $\frac{(n-1)(n-1+1)}{2} = \frac{n(n-1)}{2}$
∴ Average of n whole numbers = $\frac{n(n-1)}{n.2} = \frac{n-1}{2}$.

9. (i) Average speed = $\frac{D}{\frac{3D}{4.16}+\frac{D}{4.24}} \Rightarrow \frac{D}{\frac{3D}{64}+\frac{D}{96}}$
$= \frac{192}{11} = 17.45$ knots
(ii) Average speed = $\frac{\frac{3}{4}T \times 16 + \frac{7}{4} \times 24}{T} = \frac{18T}{T}$
= 18 knots

10. If the weight of the driver and conductor is included, it increases the average of all other passengers by 1 kg. this means they are giving 1 × 40 = 40 kg to passengers and their weight should also be considered as the new average weight.
This means the total of driver and conductor is
51 × 2 = 102 + 40 (given to passengers) = 142.
⇒ Their average = 142/2 = 71 kg.
Or, the following method can also be applied.
Their average Total weight of 40 passengers
= 40× 50=2000 kg
Total weight of 40 passengers + driver + conductor
= 42× 51 = 2142 kg.
Total weight of driver & conductor
= 2142 – 2000 = 142 kg.
Average weight of driver & conductor = $\frac{142}{2}$ = 71kg.